

Data Quality Issues Identified:

Based on the data exploration and visualizations, several data quality issues were observed:

1. Missing Values

Products Table: CATEGORY_4 has a very high proportion of missing values (778,093 missing records, ~92%). CATEGORY_3, MANUFACTURER, and BRAND also have significant missing data. The BARCODE is missing 4,025 values.

Transactions Table: BARCODE is missing 5,762 records (11.5% missing).

Users Table: LANGUAGE is missing for 30,508 users (30% missing). BIRTH_DATE, STATE, and GENDER also contain missing values.

2. Data Type Issues

Products Table: BARCODE is stored as a float instead of a string, which may cause loss of leading zeros.

Transactions Table: FINAL_QUANTITY and FINAL_SALE are stored as objects, but they should be numeric.

Users Table: Date fields (CREATED_DATE, BIRTH_DATE) are stored as objects instead of proper datetime formats.

3. Duplicates

215 duplicate rows in the Products Table.

171 duplicate rows in the Transactions Table.

No duplicates found in the Users Table.

4. Invalid or Unusual Values

Transactions Table: 8 transactions have negative barcode values, which are likely errors. 12,500 transactions have "zero" as FINAL_QUANTITY, which should be standardized to 0.

Products Table: 86,902 records have "PLACEHOLDER MANUFACTURER", indicating incomplete data.

Users Table: 9,931 records have unexpected values in the GENDER field (not "male" or "female").

Fields That Are Challenging to Understand:

1. CATEGORY_4 in Products Table:

Has 92% missing values, making it difficult to interpret.

Unclear if this category is useful for analysis.

2. FINAL_QUANTITY in Transactions Table:

Contains values like "zero", suggesting inconsistency in data entry.

3. BARCODE in Products & Transactions Tables:

Stored as a float, leading to potential loss of leading zeros.

Negative values exist, which are likely errors.

4. LANGUAGE in Users Table:

Only two main values: "en" and "es-419", but 30% of users have missing values.

"es-419" might not be immediately understandable to all analysts.

5. GENDER in Users Table:

Contains unexpected values beyond "male" and "female".

Needs further review to determine if values are valid.